



Extended observability to agentless monitoring on MySQL using ProcFS UDF plugin



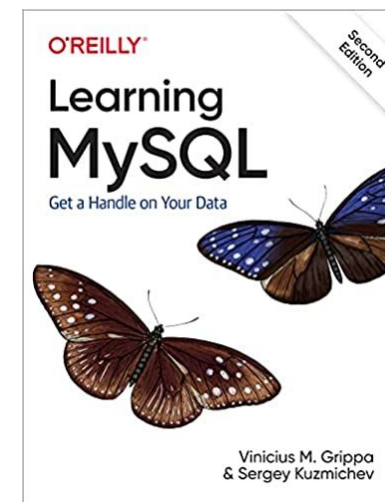
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Database Engineer at Percona

About me



- Percona Database Engineer for 6 years
- Working with databases for 18 years
- Co-author of the book Learning MySQL



Agenda

- What to monitor in a database?
- Why Monitoring?
- Agent vs Agentless Monitoring
 - Local Agent Monitoring
 - Agentless Monitoring
- ProcFS Plugin
 - Installing the plugin
 - Using the plugin
- Integrating ProcFS with PMM
- Questions

What to monitor in a database?

What To Monitor In a Database?

- Customer response time
- QPS
- Workload Behavior
- Infrastructure components (network, firewalls, storage, power supply, etc...)
- Resource utilization, security breaches
-?

Why Monitoring?

Why Monitoring?

- It helps diagnosing past and ongoing performance issues.
- It helps to identify issues proactively.
- It helps finding optimization opportunities and cost savings.
- It helps you sleep better at night.

Why Monitoring?



Agent vs Agentless Monitoring

Agent Monitoring

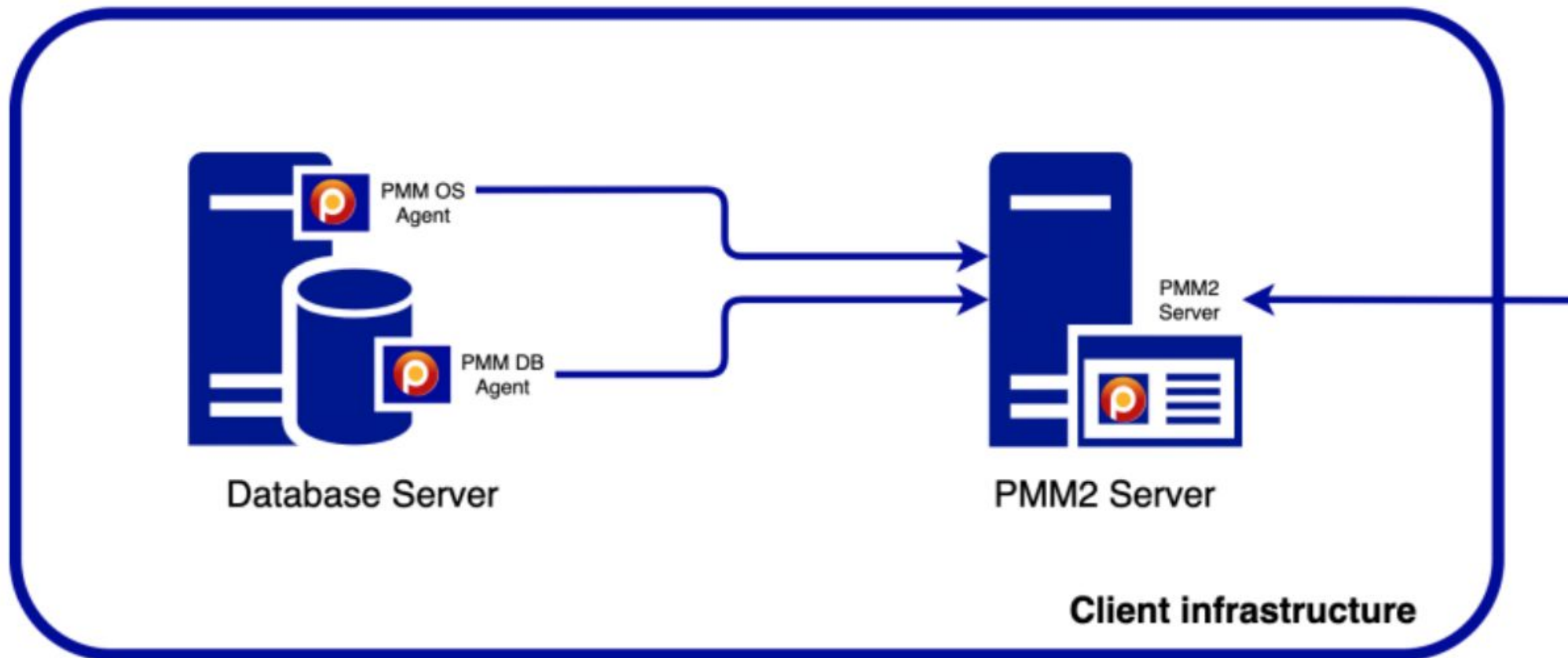
Pros:

- Greater monitoring depth is possible.
- Some agents provide a way to run custom scripts.

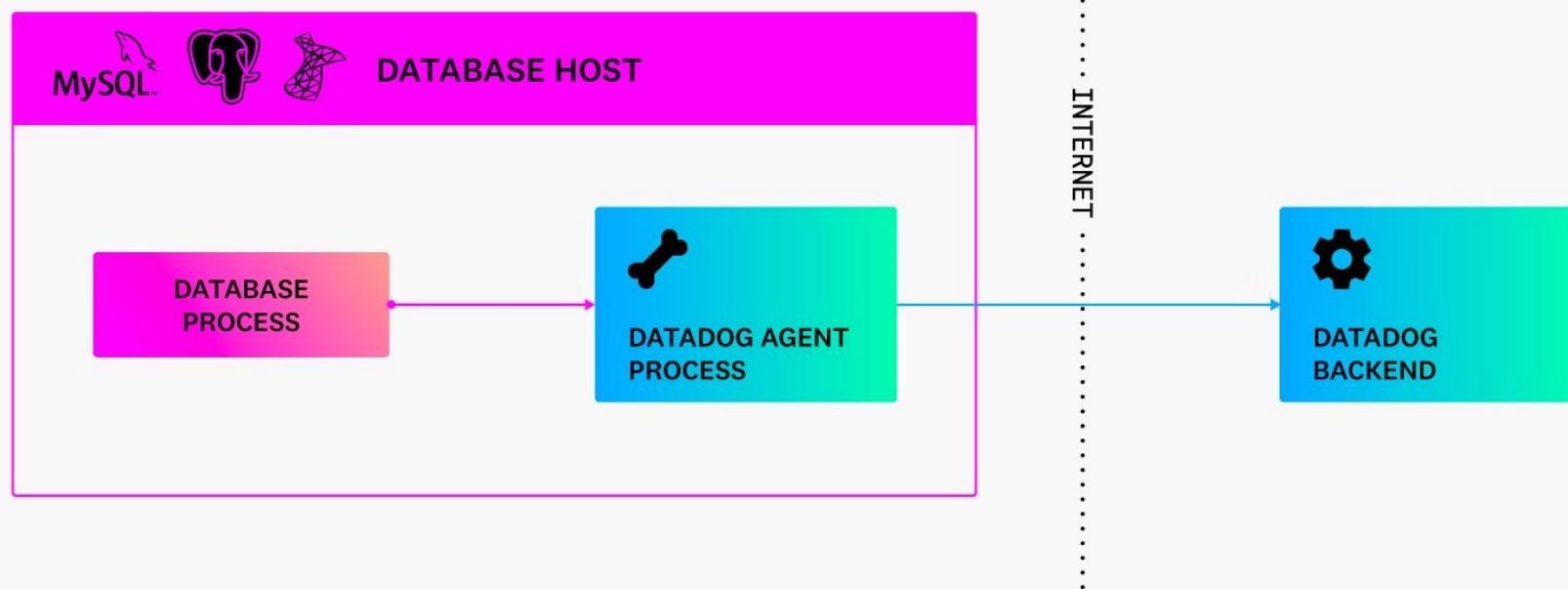
Cons:

- Agents require that you install them on your systems, so there would be an additional item to install, manage, and secure.
- Since they require something be installed, it may be more difficult to gain approval from systems teams to roll out agents.

Local Agent Monitoring



Local Agent Monitoring



Agentless Monitoring

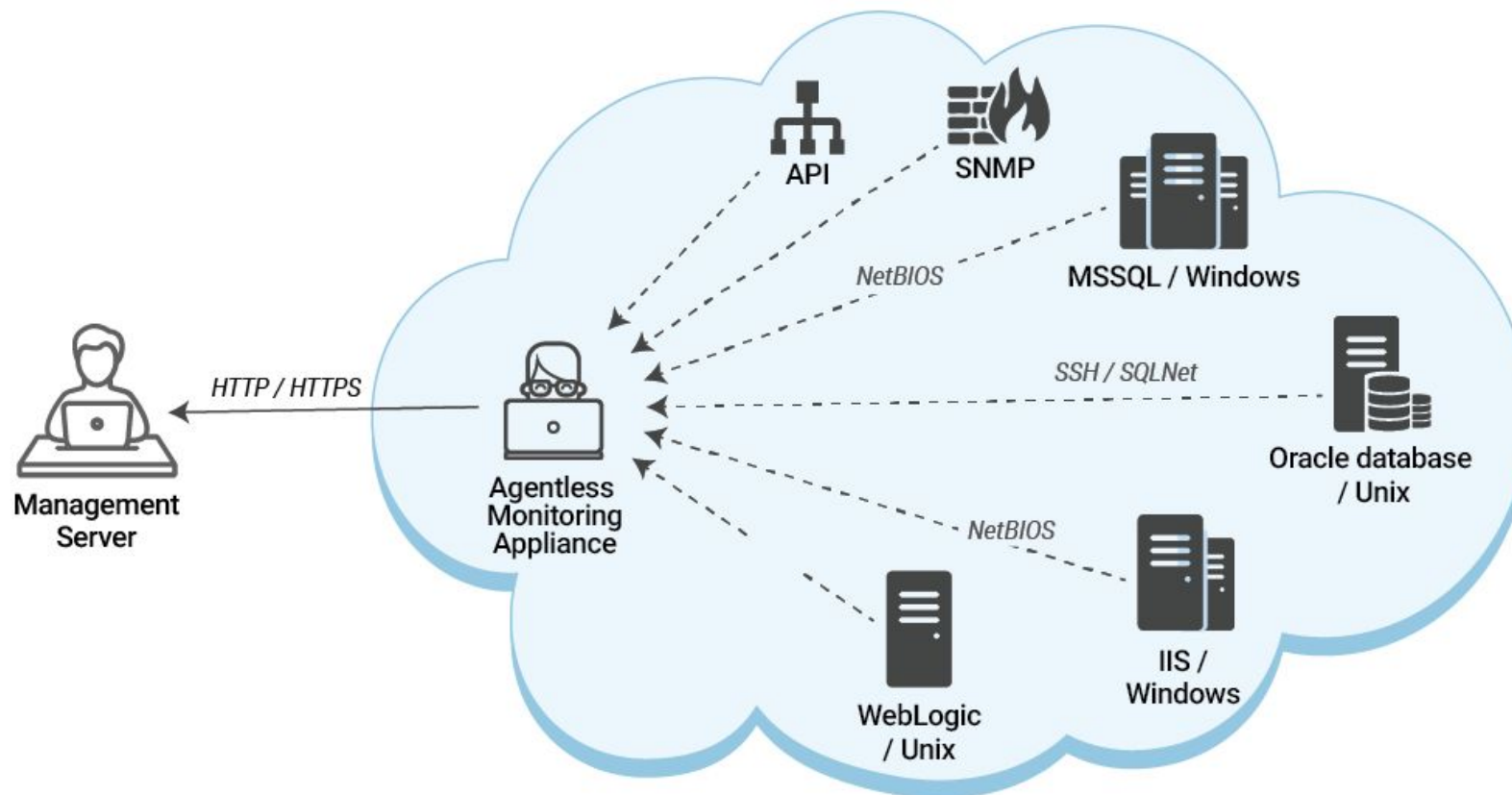
Pros:

- Agentless approaches can help reduce administrative overhead, since there is no separate agent to install or manage.
- Agentless methods may be easier to get approval to use from systems teams, since again no third-party utilities need be installed.

Cons:

- You will likely still need to take some action on the monitored host, for example, defining credentials, or loading plugins.
- The available checks is usually limited to whatever metrics are supported.
- Agentless methods can introduce higher load to the central monitoring server, so may not scale as well as agents.

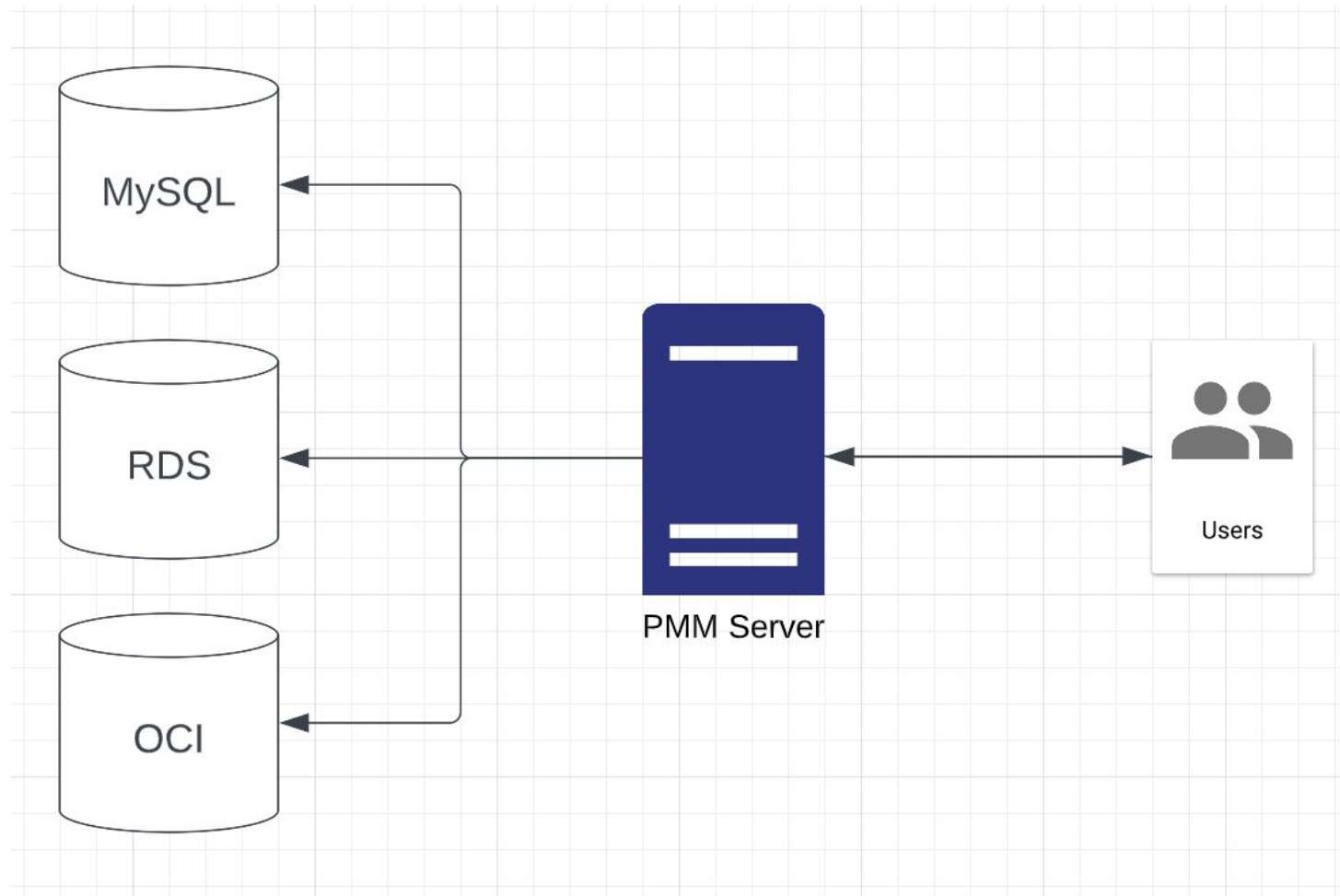
Agentless Monitoring



Agentless Monitoring of an IT infrastructure

Source: <https://www.eginnovations.com/blog/agentless-vs-agent-based-monitoring/>

Agentless Monitoring



ProcFS Plugin

ProcFS Plugin

- ProcFS plugin provides access to the Linux performance data.
- The plugin does the following:
 - Reads selected files from the `/proc` file system and the `/sys` file system.
 - Populates the file names and their content as rows in the `INFORMATION_SCHEMA.PROCFS` view.
 - User has control of what is exposed with the `procfs_files_spec` parameter(static).

ProcFS Plugin Caveats

- At the moment, it only works with Percona Server 🥲

```
18:27:48 UTC - mysqld got signal 11 ;
```

```
[...]
```

where mysqld died. If you see no messages after this, something went terribly wrong...

```
stack_bottom = 7f128441fd80 thread_stack 0x100000
```

```
[...]
```

Trying to get some variables.

Some pointers may be invalid and cause the dump to abort.

```
Query (7f123c1d9b40): SELECT * FROM INFORMATION_SCHEMA.PROCFS WHERE FILE = '/proc/version'
```

```
Connection ID (thread ID): 12
```

```
Status: NOT_KILLED
```

ProcFS Plugin Caveats

- The plugin only works within the same Percona Server version

```
mysql> INSTALL PLUGIN procfs SONAME 'procfs.so';
```

```
ERROR 1126 (HY000): Can't open shared library 'procfs.so' (errno: 0 API version for  
INFORMATION SCHEMA plugin is too different)
```

Installing ProcFS Plugin

```
mysql> INSTALL PLUGIN procfs SONAME 'procfs.so';
```

```
mysql> GRANT ACCESS_PROCFS ON *.* TO 'user'@'host';
```

Even root user needs this privilege!

```
Access denied; you need (at least one of) the ACCESS_PROCFS
```

Using ProcFS Plugin

```
mysql> SHOW GLOBAL VARIABLES LIKE 'procfs_files_spec'\G
```

```
***** 1. row
```

```
*****
```

```
Variable_name: procfs_files_spec
```

```
Value:
```

```
/proc/cpuinfo;/proc/irq/*/*//proc/loadavg;/proc/net/dev;/proc/net/sockstat;/proc/net/sockstat_rhe4;/proc/net/tcpstat;/proc/self/net/netstat;/proc/self/stat;/proc/self/io;/proc/self/numa_maps/proc/softirqs;/proc/spl/kstat/zfs/arcstats;/proc/stat;/proc/sys/fs/file-nr;/proc/version;/proc/vmstat
```

```
1 row in set (0.00 sec)
```

Using ProcFS Plugin

```
mysql> SELECT * FROM INFORMATION_SCHEMA.PROCFS WHERE FILE = '/proc/version'\G
```

```
***** 1. row *****
```

```
FILE: /proc/version
```

```
CONTENTS: Linux version 5.15.0-1028-aws (buildd@lcy02-amd64-017) (gcc (Ubuntu
```

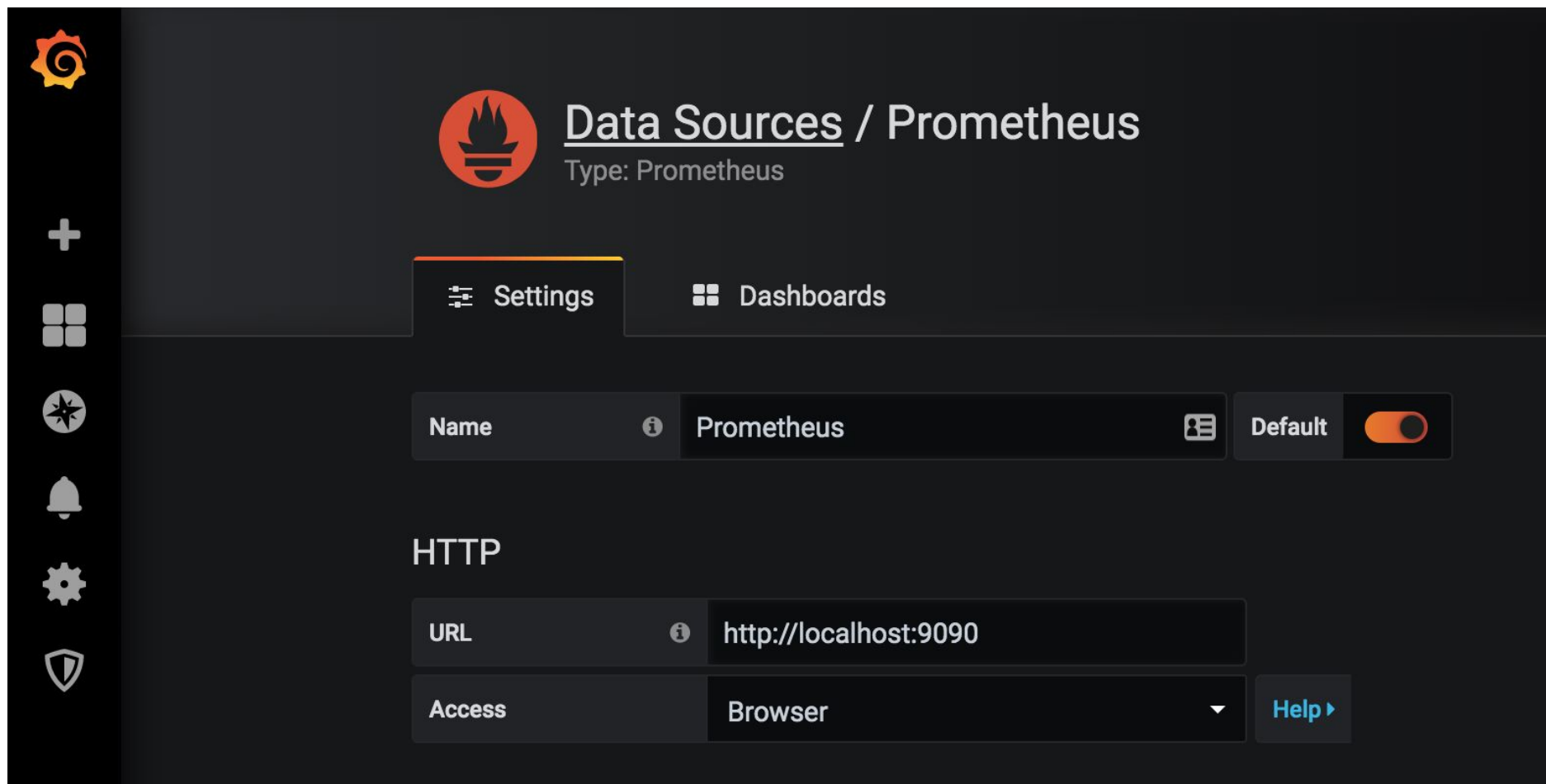
```
11.3.0-1ubuntu1~22.04) 11.3.0, GNU ld (GNU Binutils for Ubuntu) 2.38) #32-Ubuntu SMP
```

```
Mon Jan 9 12:28:07 UTC 202
```

Integrating ProcFS with Prometheus

- It is possible to integrate with Node Exporter
 - You can compile or use podman/docker
 - <https://www.percona.com/blog/procfs-udf-different-approach-to-agentless-operating-system-observability-in-your-data-base/>
 - Have `mysqlprocfs` collector
 - https://github.com/ihanick/node_exporter
- It is possible to integrate with any monitoring tool. For example with Grafana.

Integrating ProcFS with Prometheus



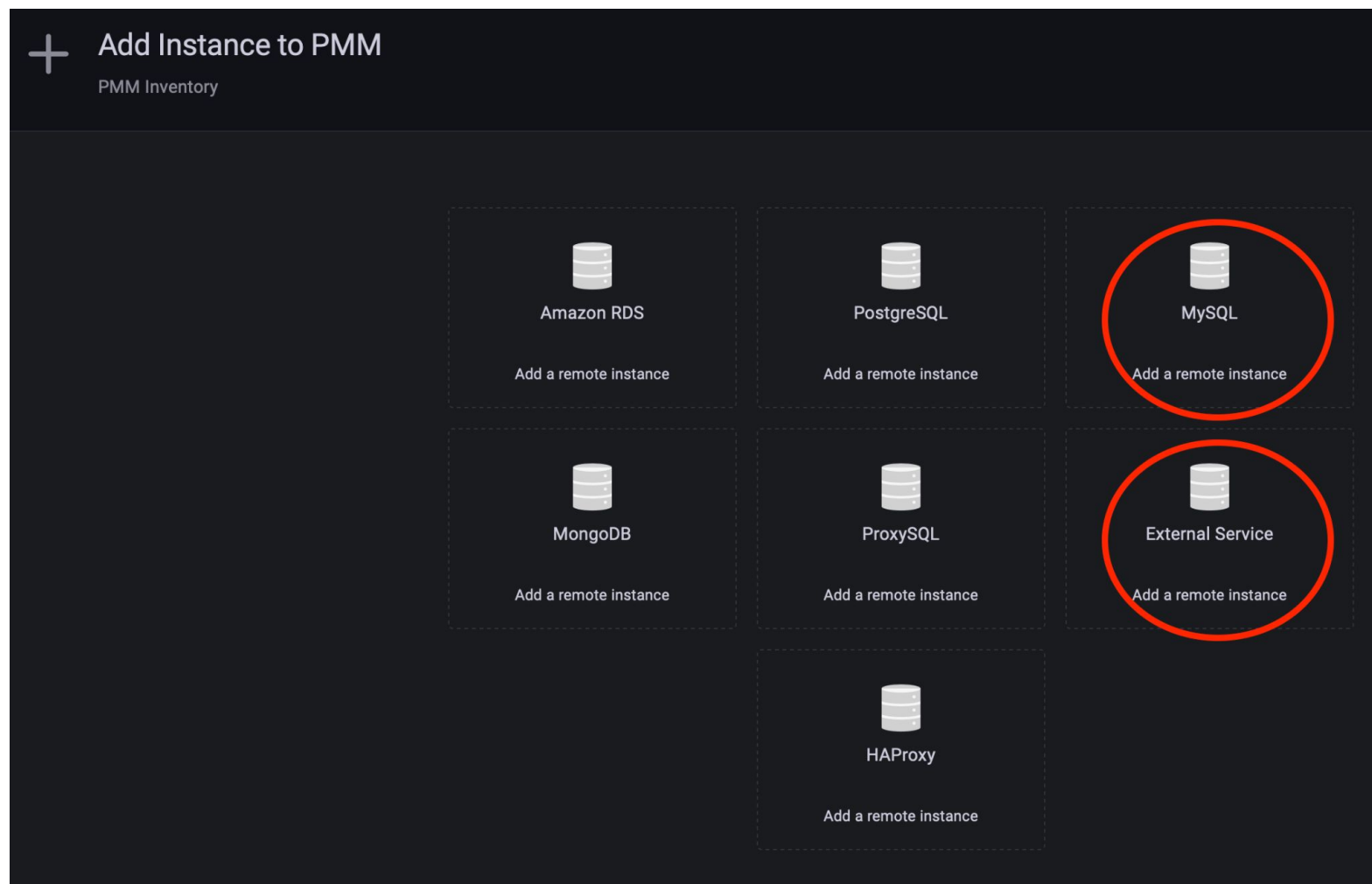
Integrating ProcFS with Prometheus

```
$ docker run -p 9100:9100 -d --name  
procfsdemo --restart always  
docker.io/perconalab/node_exporter:proc  
fs  
--collector.mysqlprocfs="vgrippa:fosdem  
2023@tcp(18.209.66.223:3306) "
```

```
$ curl 127.0.0.1:9100/metrics
```

Demo Integrating ProcFS with Grafana/PMM

Integrating ProcFS with PMM



Integrating ProcFS with PMM

ServicesAgentsNodes

Delete

	ID	Service Type	Service name	Node ID	Addresses	Port	Other Details
☐	/service_id/279a810b-bff6-4dbb-8a34-dd3a0856d9d1	MySQL	mysql_remote	/node_id/11bab135-99a1-4dd4-b699-3e191e9b448d	18.209.66.223	3306	
☐	/service_id/f8db1b87-ec64-422c-b2fe-b8d25b22d10e	PostgreSQL	pmm-server-postgresql	pmm-server	127.0.0.1	5432	database_name: postgres
☐	/service_id/f86195ea-1e78-4f34-aaa1-dc7d8201e900	External	procfsdemo	/node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28			group: external
☐	/service_id/0141182c-94e6-477b-9e18-7c0262cf9f3d	External	procfstest-node	/node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28			group: external

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28

Integrating ProcFS with PMM

```
# pmm-admin inventory list nodes
```

```
Nodes list.
```

Node type	Node name	Address	Node ID
GENERIC_NODE	ip-172-30-3-118	172.30.3.118	
	/node_id/00bfd1d-89bd-4e64-ad07-cee2d0b909a4		
GENERIC_NODE	pmm-server	127.0.0.1	pmm-server
REMOTE_NODE	mysql_remote	18.209.66.223	
	/node_id/11bab135-99a1-4dd4-b699-3e191e9b448d		
REMOTE_NODE	procfsdemo	172.17.0.3	
	/node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28		

Integrating ProcFS with PMM

```
# pmm-admin inventory add service external --name=procfstest-node  
--node-id=/node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28
```

External Service added.

Service ID : /service_id/0141182c-94e6-477b-9e18-7c0262cf9f3d

Service name : procfstest-node

Node ID : /node_id/61cbd587-7c58-4e10-8c3b-02899be0dc28

Environment :

Cluster name :

Replication set:

Custom labels : map[]

Group : external

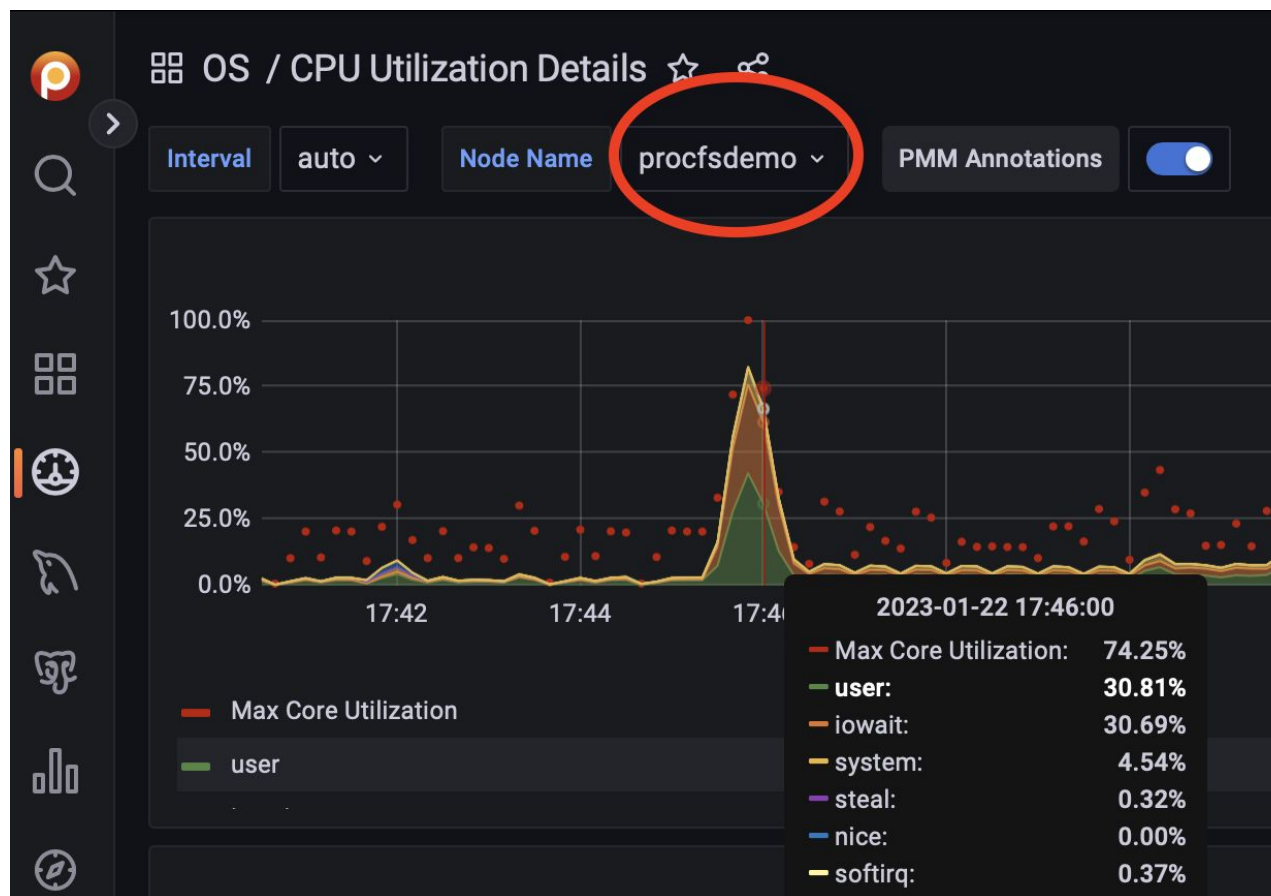
Integrating ProcFS with PMM

```
# pmm-admin inventory add agent external
--runs-on-node-id=/node_id/00bfd1d-89bd-4e64-ad07-cee2d0b909a4
--service-id=/service_id/0141182c-94e6-477b-9e18-7c0262cf9f3d --listen-port=9100
```

External Exporter added.

```
Agent ID           : /agent_id/91d6b7c8-61d6-4707-8b83-afaf25ac1db0
Runs on node ID    : /node_id/00bfd1d-89bd-4e64-ad07-cee2d0b909a4
Service ID         : /service_id/0141182c-94e6-477b-9e18-7c0262cf9f3d
Username           :
Scheme             : http
Metrics path       : /metrics
Listen port        : 9100
Disabled           : false
Custom labels      : map[]
```

Integrating ProcFS with PMM



Questions?

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谢谢
Thank you
Grazie
Obrigado
Gracias

Installing ProcFS Plugin

```
mysql> INSTALL PLUGIN procfs SONAME 'procfs.so';
```

```
mysql> GRANT ACCESS_PROCFS ON *.* TO 'user'@'host';
```

Even root user needs this privilege!

```
Access denied; you need (at least one of) the ACCESS_PROCFS
```

```
> SELECT * FROM INFORMATION_SCHEMA.PROCFS WHERE FILE = '/proc/version' \G
1227 (42000): Access denied; you need (at least one of) the ACCESS_PROCFS privi
> select user,host from mysql.user;
```

Integrating ProcFS with Prometheus

- Prerequisites
 - Have pmm-client installed and configured
 - Have a node exporter running (you can compile or use podman/docker) –

<https://www.percona.com/blog/procfs-udf-different-approach-to-agentless-operating-system-observability-in-your-data-base/>